

Stochastic Mathematics for Finance

Week 3 Assignment

Brownian Motion, Itô's Lemma, and Stochastic Differential Equations

Instructions

- This assignment contains 3 computational and theoretical case studies in stochastic calculus.
- Each question contains both a theoretical component and a computational component.
- For computational parts, submit the relevant Python code and a short explanation of your observations.
- You are encouraged to visualize your simulations wherever appropriate.
- Briefly interpret the meaning of your numerical and simulation results.
- You may use the Week 3 resources, but some parts require applying the ideas in a new setting.

Submission Format

Submit your work on GitHub in a clean repository containing:

- a README.md file with your name, roll number, and GitHub username;
- a single PDF named solutions.pdf;
- Python code files inside a folder named code/, named clearly as q1.py, q2.py, etc.;
- plots or generated outputs, if any, inside a folder named outputs/.

Question 1: Brownian Motion Paths and Quadratic Variation

A quantitative analyst is studying the path properties of Standard Brownian Motion to build intuition for stochastic calculus.

Let $\{W_t\}$

Let $\{W_t\}_{t \geq 0}$ be a Standard Brownian Motion. Recall that for any $0 \leq s < t$ the increment $W_t - W_s$ is independent of all information up to time s and satisfies:

$$W_t - W_s \sim N(0, t - s).$$

(a) State the four defining properties of Standard Brownian Motion. For each property, give a brief one-sentence interpretation in financial terms.

(b) Derive the covariance formula

$$\text{Cov}(W_s, W_t) = \min(s, t) \quad \text{for all } s, t \geq 0.$$

Your proof must explicitly use the independent increments property and should not assume the result.

(c) Define the quadratic variation Q^n of a stochastic process on $[0, T]$ over a partition $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$. Show that

$$E[Q^n] = T, \quad \text{where } Q^n = \sum_k (W_{t_{k+1}} - W_{t_k})^2.$$

(d) Show that $\text{Var}(Q^n) \rightarrow 0$ as $\|\Pi\| \rightarrow 0$, and conclude that $Q^n \rightarrow T$ in L^2 . Explain in one or two sentences why it is remarkable that the quadratic variation of a random path converges to a deterministic quantity.

(e) Using Python, simulate a single Brownian path on $[0, 1]$ with $N = 10,000$ steps. Compute the empirical quadratic variation of the path and compare it to the theoretical value $T = 1$. Also compute the total variation of the same path and comment on the contrast.

(f) Plot the empirical quadratic variation as a function of the number of partition steps N , for

$$N \in \{10, 50, 100, 500, 1000, 5000, 10000\}.$$

Overlay the theoretical value $T = 1$ as a horizontal dashed line. Comment on the rate of convergence.

(g) Estimate the empirical mean and variance of W_t at a fixed time $t = 0.5$ using 10,000 simulated paths. Compare to the theoretical values $E[W_t] = 0$ and $\text{Var}(W_t) = t$.

Hint

For part (b), write $W_t = W_s + (W_t - W_s)$ and use the fact that W_s and $(W_t - W_s)$ are independent.

For part (d), use the fact that for a $N(0, \sigma^2)$ variable Z , we have $\text{Var}(Z^2) = 2\sigma^4$.

For part (f), the standard deviation of the estimator Q^n is of order $O(\sqrt{\Delta t})$, so coarser partitions are noisier.

In Python: `increments = np.diff(W); QV = np.sum(increments**2); TV = np.sum(np.abs(increments))`.

Question 2: Itô's Lemma and Geometric Brownian Motion

A derivatives desk needs to derive and numerically validate the explicit solution to the Geometric Brownian Motion SDE using stochastic calculus.

Suppose the stock price S_t satisfies:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 = 100,$$

with drift $\mu = 0.07$ and volatility $\sigma = 0.2$.

(a) State Itô's Lemma for $f(t, X_t)$ where X_t satisfies $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$. Identify each term in the result and explain in one sentence why the ordinary chain rule must be modified for stochastic processes.

(b) Apply Itô's Lemma to $f(S_t) = \ln(S_t)$ and show that

$$d(\ln S_t) = (\mu - \sigma^2/2) dt + \sigma dW_t.$$

Hence integrate to derive the explicit solution:

$$S_t = S_0 \exp\{(\mu - \sigma^2/2)t + \sigma W_t\}.$$

(c) Explain what the Itô correction term $-\sigma^2/2$ represents. Compute $E[S_t]$ using the explicit solution and show it equals $S_0 e^{\mu t}$. Compute the incorrect expected value that results from omitting the correction and quantify the error for $\sigma = 0.2$ and $t = 10$ years.

(d) Using Python, simulate 50,000 paths of S_t via the Euler–Maruyama method with $T = 1$ year and $N = 500$ steps. Plot a histogram of the terminal stock prices S^1 and overlay the theoretical log-normal density.

(e) Report the empirical mean and standard deviation of S^1 and compare them to the theoretical values:

$$E[S^1] = S_0 e^{\mu T}, \quad Std[S^1] = S_0 e^{\mu T} \sqrt{(e^{\sigma^2 T} - 1)}.$$

(f) Repeat the simulation for $\sigma \in \{0.1, 0.3, 0.5\}$ keeping $\mu = 0.07$ fixed. For each value of σ , report the empirical mean and standard deviation of S^1 . Briefly explain how volatility affects:

- the spread of the terminal stock price distribution;
- the expected option payoff $E[(S^1 - K)^+]$ for $K = 110$.

(g) The log-return over $[0, T]$ is defined as $\ln(S^1/S_0)$. Using your simulations, verify empirically that log-returns are approximately normally distributed and estimate their mean and variance. Compare to the theoretical values $(\mu - \sigma^2/2)T$ and $\sigma^2 T$.

Hint

For the Itô correction in part (c): $E[e^{\sigma W_t}] = e^{\sigma^2 t/2}$ for $W_t \sim N(0, t)$, which follows from the MGF of a Gaussian.

The Euler–Maruyama step for GBM is: $S_{k+1} = S_k + \mu S_k \Delta t + \sigma S_k \Delta W_k$, where $\Delta W_k \sim N(0, \Delta t)$.

In Python: `dW = np.random.normal(0, np.sqrt(dt), size=(n_paths, N))`

The theoretical log-normal density can be overlaid using `scipy.stats.lognorm`.

Question 3: Euler–Maruyama Simulation and the Ornstein–Uhlenbeck Process

A fixed income team wants to model short-term interest rates using a mean-reverting stochastic process that does not allow explosive growth.

The Ornstein–Uhlenbeck (OU) process is defined by the SDE:

$$dX_t = -\theta X_t dt + \sigma dW_t, \quad X_0 = x_0,$$

with mean-reversion speed $\theta > 0$ and diffusion coefficient $\sigma > 0$.

(a) Identify the drift and diffusion coefficients of the OU SDE. Explain in financial terms why the drift term $-\theta X_t dt$ causes mean reversion toward zero, and what happens to the drift when X_t is very large or very negative.

(b) Write down the Euler–Maruyama discretisation of the OU SDE with time step Δt :

$$X_{tk+\Delta t} = X_{tk} - \theta X_{tk} \Delta t + \sigma \Delta W_k.$$

State the stability condition on Δt for the scheme to remain well-behaved.

(c) The stationary distribution of the OU process satisfies

$$X^\infty \sim N(0, \sigma^2 / 2\theta).$$

Explain what this means: in what sense does the process ‘converge’ to this distribution, and why is a stationary distribution useful in interest rate modelling?

(d) Using Python, simulate 10 paths of the OU process with parameters $X_0 = 3.0$, $\theta = 2.0$, $\sigma = 0.5$, $T = 5$, $N = 5000$ steps. Plot all paths on the same axes, adding:

- a dashed horizontal line for the stationary mean (zero);
- a shaded band for $\pm 2\sigma/\sqrt{2\theta}$ (the ± 2 standard deviation range of the stationary distribution).

(e) Verify the stationary distribution empirically: simulate 20,000 independent paths with the same parameters but $T = 10$. Record the terminal value X^1 for each path and plot a histogram. Overlay the theoretical $N(0, \sigma^2/(2\theta))$ density. Report the empirical mean and variance and compare to theory.

(f) Repeat the path simulation for $\theta \in \{0.5, 2.0, 5.0\}$ keeping $\sigma = 0.5$ and $X_0 = 3.0$ fixed. Plot the paths for each value of θ on separate subplots. Briefly explain how θ affects:

- the speed at which paths return toward zero;
- the width of the stationary distribution.

(g) The Euler–Maruyama scheme has strong convergence order $1/2$ and weak convergence order 1 . Explain in plain terms what strong and weak convergence mean, and why weak convergence is sufficient for estimating $E[g(X^1)]$ in option pricing applications.

Hint

The stability condition for the OU Euler–Maruyama scheme is $\Delta t < 2/\theta$.

For the stationary variance: $\sigma^2/(2\theta)$. For $\theta = 2.0$, $\sigma = 0.5$ this gives 0.0625 , so $\text{std} \approx 0.25$.

In Python: `dW = np.random.normal(0, np.sqrt(dt)); X[k+1] = X[k] - theta*X[k]*dt + sigma*dW.`

For part (e), use a long horizon $T = 10$ so the distribution has time to reach stationarity.

`scipy.stats.norm.pdf` can overlay the theoretical Gaussian density on your histogram.

General Notes

Throughout this assignment, focus on:

- understanding the path properties of Brownian motion and how they differ from smooth functions;
- interpreting the Itô correction and its financial consequences;
- comparing theoretical quantities (means, variances, distributions) with numerical estimates;
- visualizing stochastic processes and their long-run behaviour computationally.

The Euler–Maruyama method and the SDE framework studied here underpin virtually every continuous-time model in quantitative finance, from Black–Scholes to stochastic volatility and interest rate models. This assignment is intended to provide intuition for how these models behave in practice.